

Problem 1

Discuss the behavior of a Nadaraya-Watson mean regression estimate when one of i.i.d. observations, say (x_1, y_1) , tends to assume extreme values. Specifically, discuss

- (a) the case when x_1 is very big,
- (b) the case when y_1 is very big.

Problem 2

Firms produce some product using technology $f(l, k)$. The functional form of f is unknown, although we know that it exhibits constant returns to scale (CRS). For a firm i , we observe labor l_i , capital k_i , and output y_i . The data generating process takes the form

$$y_i = f(l_i, k_i) + \varepsilon_i$$

where $\mathbb{E}[\varepsilon_i] = 0$, and ε_i is independent of (l_i, k_i) .

Using random sample $\{(y_i, l_i, k_i)\}_{i=1}^n$ suggest a nonparametric estimator of $f(l, k)$ which also exhibits CRS.

Problem 3

Recall the Nadaraya-Watson kernel estimator $\hat{g}(x)$ of the conditional mean $g(x) := \mathbb{E}[y|x]$ constructed for a random sample. Show that if $g(x) = c$, where c is some constant, then $\hat{g}(x)$ is unbiased.

Problem 4

Suppose we have a random sample $\{x_i\}_{i=1}^n$ and let

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \leq x\}$$

denote the empirical distribution function of x_i . Consider two density estimators:

- one-sided estimator:

$$\hat{f}_1(x) = \frac{\hat{F}(x+h) - \hat{F}(x)}{h},$$

- two-sided estimator:

$$\hat{f}_2(x) = \frac{\hat{F}(x+h/2) - \hat{F}(x-h/2)}{h}.$$

Show that:

- (a) $\hat{F}(x)$ is an unbiased estimator of $F(x)$.
- (b) The bias of $f_1(x)$ is $O(h^a)$. Find the value of a .
- (c) The bias of $f_2(x)$ is $O(h^b)$. Find the value of b .

Problem 5

Derive the asymptotic distribution of the Nadaraya-Watson estimator of the density of a scalar random variable x having a continuous distribution, similarly to how the asymptotic distribution of the Nadaraya-Watson estimator of the regression function is derived, under similar conditions. Give interpretation to how your expressions for asymptotic bias and asymptotic variance depend on the shape of the density.